

Electromagnetic Spectrum Operations Characteristics of Need Base Content

1. **Situation.** The rapid development of electromagnetic spectrum dependent technologies has fundamentally reshaped modern EMSO. Achieving EMSO dominance requires capabilities to sense, locate, attack, and defeat enemy spectrum-dependent systems while guaranteeing the commander's uninterrupted use of the EMS to achieve objectives. Commanders must employ EMSO capabilities that enable decision advantage against a peer adversary in the Denied, Degraded, Intermittent Connectivity, or Limited Bandwidth (DDIL) battlefields of Multi-Domain Operations.

2. **Problem.** The Army lacks the ability to sense, locate, attack, and protect in and through the EMS across competition and geographic continuums in order to generate necessary effects against targets of interest at range. Currently, EMSO capabilities are spread across different warfighting functions and not fully designed as cohesive technologies that are modular, scalable, and adaptable enough to mitigate modern threats. This prevents the Army from truly leveraging Artificial Intelligence and Machine Learning (AI/ML) for quick decision making to exploit opportunities across the competition continuum. Dominance in the EMS will depend on our ability to effectively attack, protect, support and manage use of the spectrum and related EMSO common services to create and maintain an advantage across multiple domains and across all warfighting functions.

3. **Objective.** At its core, the EMSO modernization effort is the continuous evolution of flexible requirements, industry partnerships, and governance. Delivering EMSO capabilities requires innovation across warfighting function requirements, resourcing, acquisition and contracting to drive capability and enable an advantage in four key areas:

- a. **Attack.** Modern EMSO systems must provide commanders with the ability to affect current and emerging spectrum dependent systems at the necessary distances using a variety of platforms including, but not limited to, aircraft, ground sites, maritime vessels, space assets, and cyberspace. Systems must provide a full range of electromagnetic attack (EA) techniques that include noise jamming, protocol-based effects, and other methods that deny, degrade, disrupt, destroy, or manipulate the adversary's ability to communicate, sense, control, and navigate using the EMS.
- b. **Support.** Modern EMSO systems must provide comprehensive sensing capabilities to detect, identify, and locate target communication and non-communication emitters in the operational environment. They must simultaneously perform identification and direction finding against multiple emitters, receive and process signals from various sources, and provide real-time tactical information that enable situational awareness/understanding, targeting, decision making, and planning cycles.

- c. **Protect.** Personnel, facilities, and equipment must be protected from the effects of adversary EA to ensure authorized use of the spectrum. Measures include hardening, emission control, reprogramming (e.g. development of electromagnetic countermeasures), EMS manipulation, and obfuscation capabilities that alter electromagnetic signatures to hide their original meaning, thereby providing a layer of protection against reverse engineering, breaches, and unauthorized access.
- d. **Common Services.** Modern EMSO systems must provide core shared functions to integrate within common open architecture frameworks. A foundation built on data exchange standardization will enable rapid onboarding of new technologies and third-party applications to include sustained common user training with simple and intuitive graphic user interfaces.

4. **Key Characteristics.** The core concept of future EMSO capabilities is an open and modular ecosystem of EMS dependent systems that can be employed on any platform. This system-of-system approach provides the commander with flexible options to attack, sense, and protect the EMS at echelon. Army EMSO systems must be secure, interoperable across warfighting functions and fully integrated with the Army's Next Generation Command and Control (NGC2) architecture. This integration will ensure robust and real-time utilization of the NGC2 technology stack.

- a. **Resilient.** The Army needs modern EMSO systems that facilitate spectrum management and protect friendly forces from the effects of adversary EA and ensure their own unimpeded access to and use of the electromagnetic spectrum. Measures must include hardening as well as electromagnetic countermeasures to ensure the resilience and reliability of friendly systems.
- b. **Adaptable.** The Army must build modern EMSO capabilities on software-defined architecture. This enables reprogramming updates to be sent to the systems sensors and effectors to counter adversary signals. EMSO systems must also be capable of supporting an array of swappable multi-wave/multi-beam antennas to ensure they can operate at range and direction against ever-changing threats.
- c. **Modular.** The Army must possess EMSO systems that adhere to Modular Open Systems Approach (MOSA) and Sensor Open Systems Architecture (SOSA) standards to allow for removal or replacement of system components and/or platforms to meet mission needs. Modularity is achievable with capabilities that stay within size, weight, and power constraints.
- d. **Scalable.** The Army must be able to scale EMSO systems to account for changes in amplifiers, antennas, and platforms needed to prosecute all targets at echelon.
- e. **Empowered.** Army EMSO systems must incorporate AI/ML that enables quicker

decision making for target sensing, Direction Finding (DF), geolocation (GEO), effect pairing, and delivery.

- f. **Rapidly Deployable.** The Army must design EMSO systems for immediate adoption, and quick usability enabling swift transportation, minimal setup time, and rapid integration into diverse tactical environments. Deployability is also enhanced through the use of systems that can be manned/unmanned and mounted/dismounted across land, air, maritime, and space.
 - g. **Secure.** Army EMSO systems must be capable of receiving secure updates, including rekeying material, over-the-air to ensure cryptographic integrity and continued secure operations.
 - h. **Interoperable.** Army EMSO capabilities must be designed with consideration of an open-architecture framework enabling it to operate within the Next Generation Command and Control (NGC2) framework. Systems must be able to ingest, process, and share data with a wide array of command-and-control, intelligence, and fires networks.
5. **Amendments.** This Characteristics of Need statement is a living document and will be updated as required based on technological observations and lessons learned through experimentation.